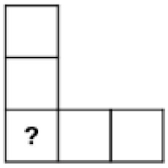
START for ALL PARTICIPANTS

1. SETSQUARE (coefficient 1)

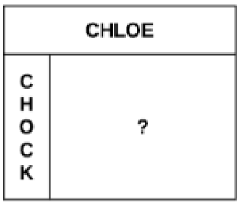
Write all the whole numbers from 1 to 5 in the boxes, one number per box. The sum of the three numbers in the horizontal row and the sum of the three numbers in the vertical column must equal 9.



What number will be written in place of the question mark?

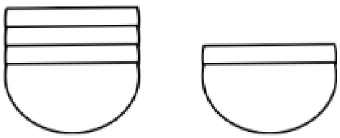
2. CHOCOLATE BAR (coefficient 2)

A Swiss chocolate bar is shaped like a rectangle, composed of identical squares. Chloe takes a full horizontal strip that goes from the left edge to the right edge of the bar. She eats the 5 squares thus obtained. Then Chock takes a full vertical strip that runs from top edge to bottom edge from what's left of the bar. He eats the 3 squares thus obtained.



How many squares are finally left?

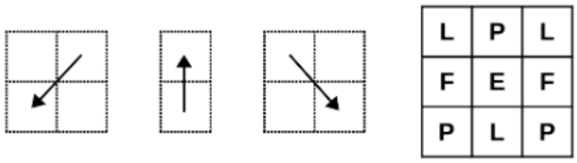
3. BOWLS (coefficient 3)



The bowls are identical. In height, a stack of four bowls is 13 centimetres and a stack of two bowls is 9 centimetres.

How tall would a stack of six bowls be in centimetres?

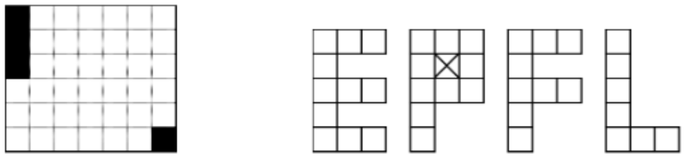
4. PATHS (coefficient 4)



Maisie can move from one box to another in each of the three directions indicated.

How many paths allow her to read the EPFL acronym on the board?

5. LOGO (coefficient 5)



For the final of the championship, Valerie would like to make a banner in honour of EPFL. She wants to cut the four letters out of a rigid rectangular plate divided into a 6 x 7 grid. Unfortunately, the four squares in black in the drawing are damaged. As the plate is painted on only one face, the face visible in the drawing, Valerie can turn but not turn over any letter template. Indeed, the correct face of each letter, the one that is readable, must be in colour on the banner. She must place each letter template on the plate so that it does not cover any damaged squares and, of course, so that it does not overlap any other letter template.

Trace the outline of the four letters on the plate.

END for CE PARTICIPANTS

6. THEATRE SEATING (coefficient 6)

In the theatre, seven friends are seated next to each other in the same row. From the central aisle, their seats are numbered from 1 to 7. During the interval, Amber gets up to go to the refreshment bar. On the way back, she notices that Babs and Carole have moved closer to the aisle, one by four seats and the other by two seats, while Denise and Emma have moved away from the aisle, one by three seats and the other by one seat. She also notices that Fabienne and Grace have swapped seats. The resultant free seat, in grey, is numbered 4.

What was Amber's seat number before the interval?

7. MAGIC SQUARE 36 (coefficient 7)

	6		
4		10	
	1		?
7		15	

Sixteen strictly positive integers, all different, must be written in the boxes, one number per box. The sum of the four numbers in each of the four columns, in each of the four rows and in each of the two diagonals must equal 36. Six numbers are already written.

What number will be written in place of the question mark?

8. THE PIRATES (coefficient 8)

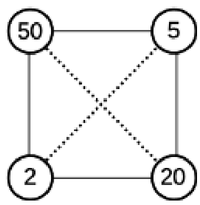
The crew of a pirate ship shares 36 gold coins and 84 silver coins. The crew is comprised of different ranks: commanders, masters, sailors and gunners. Each commander receives four gold coins and six silver coins. Each master receives two gold coins and four silver coins. Each sailor receives one gold coin and three silver coins. Similarly, each gunner receives one gold coin and three silver coins.

How many people are there in the crew of the ship?

END for CM PARTICIPANTS

Problems 9 to 18: beware! For a problem to be completely solved, you must give both the number of solutions, AND give the solution if there is only one, or give any two correct solutions if there are more than one. For all problems that may have more than one solution, there is space for two answers on the answer sheet (but there may still be just one solution).

9. SQUARES (coefficient 9)



Four strictly positive integers, all different, must be written at the vertices of a square, one number per vertex. If two numbers are on the same side, one must be a multiple of the other. If two numbers are on the same diagonal, dotted on the drawing, neither can be a multiple of the other. In the drawing, the sum of the four numbers is 77 and the largest number is 50 (each number is written inside a disc).

For another square, what is the sum of the four numbers if the largest number is 36?

10. NINE LETTERS, EIGHT “WORDS” (coefficient 10)

This is a two-player game, with nine cards bearing the letters A, D, E, G, I, N, O, S and T, one letter per card. At the start of a game, the nine cards are placed, letter face up, on the table and each player's display is empty. In turn, each player chooses a letter from the table and places it on their display. A player wins when the letter they has just drawn and two letters that were already on their display allow them to form one of the eight combinations (“words”) ADO, AGE, AIT, DIS, EST, GIN, OIE or ONT. A game can be drawn, that is to say end in the ninth round without a winner or loser. Caroline plays first and chooses the letter A. Sebastian chooses the letter D in the second round. Caroline chooses the letter O in the third round.

Which letter must Sebastian choose in the fourth round so that Caroline cannot win in the seventh round?

11. THREE CREATURES (coefficient 11)

In this equation, one letter always represents the same digit and two different letters always represent two different digits.

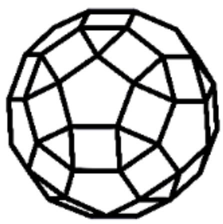
PEPE + ELLE + FEE = ELFE + ELFE + ELFE

What number is represented by EPFL?

Note: multi-digit numbers never start with 0, e.g. the answer 1306 is not accepted.

END for C1 PARTICIPANTS

12. POLYHEDRON OF THE YEAR (coefficient 12)



Two squares alternate with an equilateral triangle and a regular pentagon around each vertex of a semi-regular polyhedron. The number 36 appears on each triangular or pentagonal face. A certain number, always the same, appears on each square face. The sum of the numbers on all the faces is equal to 2022.

What number appears on each square face?

13. SQUARE TABLES (coefficient 13)

Strictly positive integers are written in the cells of a square table, one number per cell, with at minimum four cells in total. The same number can appear several times in a row or column. The sum of the numbers in each row or column is always the same. While programming, Daniel counts 2021 different tables if the sum of all the numbers in the table equals 28.
By hand, Jean-Louis counts 11, 21 and 63 tables respectively if this total sum is equal to 12, 15 and 18.

How many tables are possible if the sum of all the numbers is equal to 21?

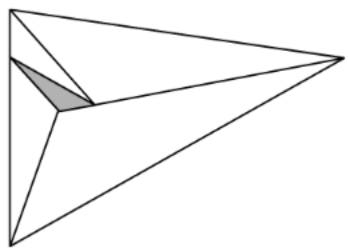
14. PAULA'S PRECISE PAVER (coefficient 14)

Each of the three dimensions of Paula's rectangular parallelepiped paver is a strictly positive integer. The total area, necessarily even, cannot take certain values, such as 36. As examples: it is equal to 100 if the dimensions are 1, 2 and 16, and to 200 if the dimensions are 2, 2 and 24.

What even integer, between 100 and 200, cannot be the total area of the six faces?

END for C2 PARTICIPANTS

15. FATHER AWAY'S PENNANT (coefficient 15)

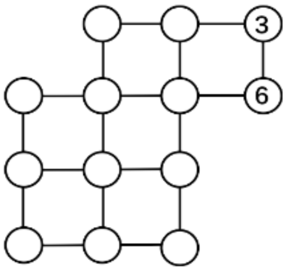


Father Away had sewn a pennant from five unequal pieces of fabric and hoisted it to the mast of his life raft. The pennant is sewn as five similar triangles. The area of the smallest triangle, in grey, is 36 cm^2.

What is the area of the pennant, in cm^2 rounded to the nearest cm^2?

Note: in a triangle, if the angle between two sides a and b is 120°, the third side c is given by: $c^2 = a^2 + b^2 + ab$.

16. ONE TO THIRTEEN (coefficient 16)

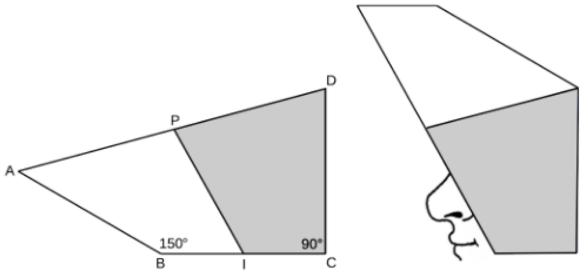


Write all the integers from 1 to 13 to the discs, one number per disc.

The sum of the four numbers written in the discs at the vertices of each of the six small squares must be equal to 25. The numbers 3 and 6 are already written.

END for L1, GP PARTICIPANTS

17. MEDIEVAL HELMET (coefficient 17)



The drawing represents the cross-section of a medieval helmet from the 14th century. The quadrilateral PABI, in white, corresponds to a visor which pivots around P. Its area is 182 cm^2. The interior angle IBA measures 150°. The quadrilateral PICD, in grey, corresponds to a protective shell. The interior angle DCI measures 90°. I and P are respectively the midpoints of the segments [BC] and [DA]. The lengths AB, BC and CD are equal.

What is, in cm^2 rounded to the nearest cm^2, the area of the quadrilateral PICD?

Note: take $\sqrt{3}$ as 1.732.

18. STAMPS (coefficient 18)

Expressed in the official currency, the prices of the 4 stamps issued by the postal service of Mathsland are 4 strictly positive integers, all different, whose sum is equal to 33. On an envelope, one can affix several stamps, including those of identical price. By using no more than 4 stamps on an envelope, one can pay any postage from 1 to 44 inclusive.

What are, in ascending order, the prices of the 4 stamps issued by the postal service?

END for L2, HC PARTICIPANTS